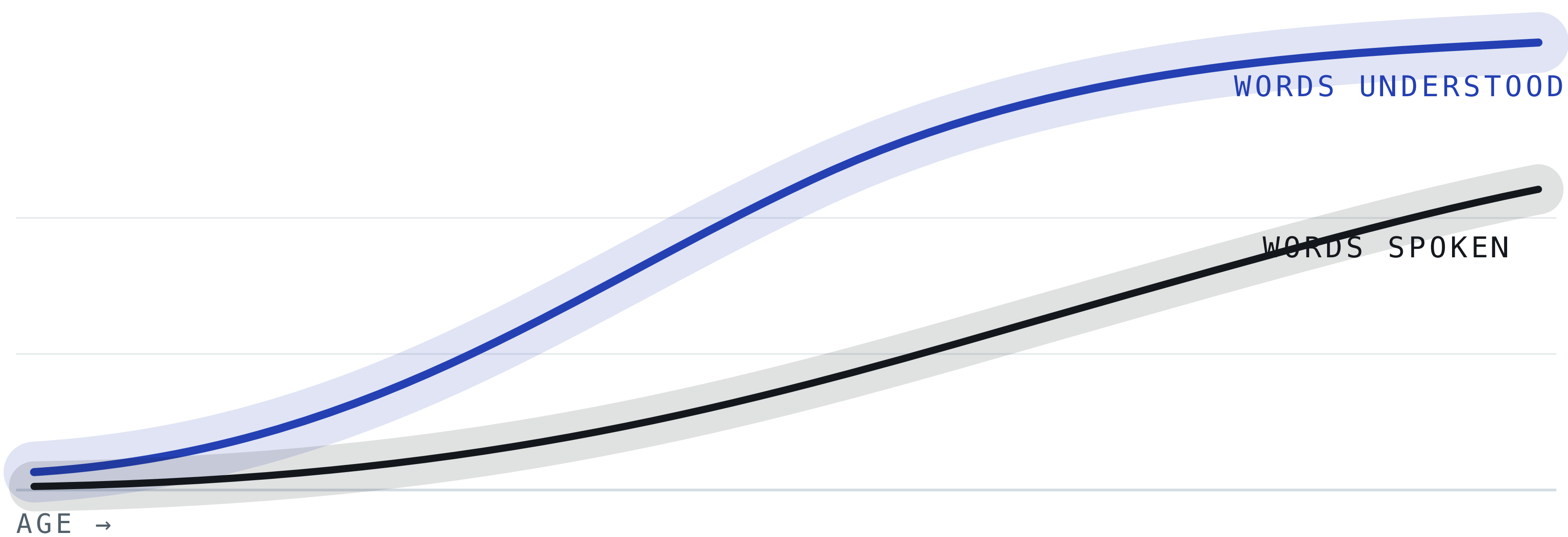
Down Syndrome Education International · Vocabulary Growth Study

How children with Down syndrome build vocabulary

A pooled Bayesian study of the words children understand, say and sign, across fourteen datasets from six countries.

1,424 assessments · 14 datasets · ages 8 months to 9½ years · production ratio reported to 6 years



Schematic · growth curves with credible bands · the fitted versions are inside

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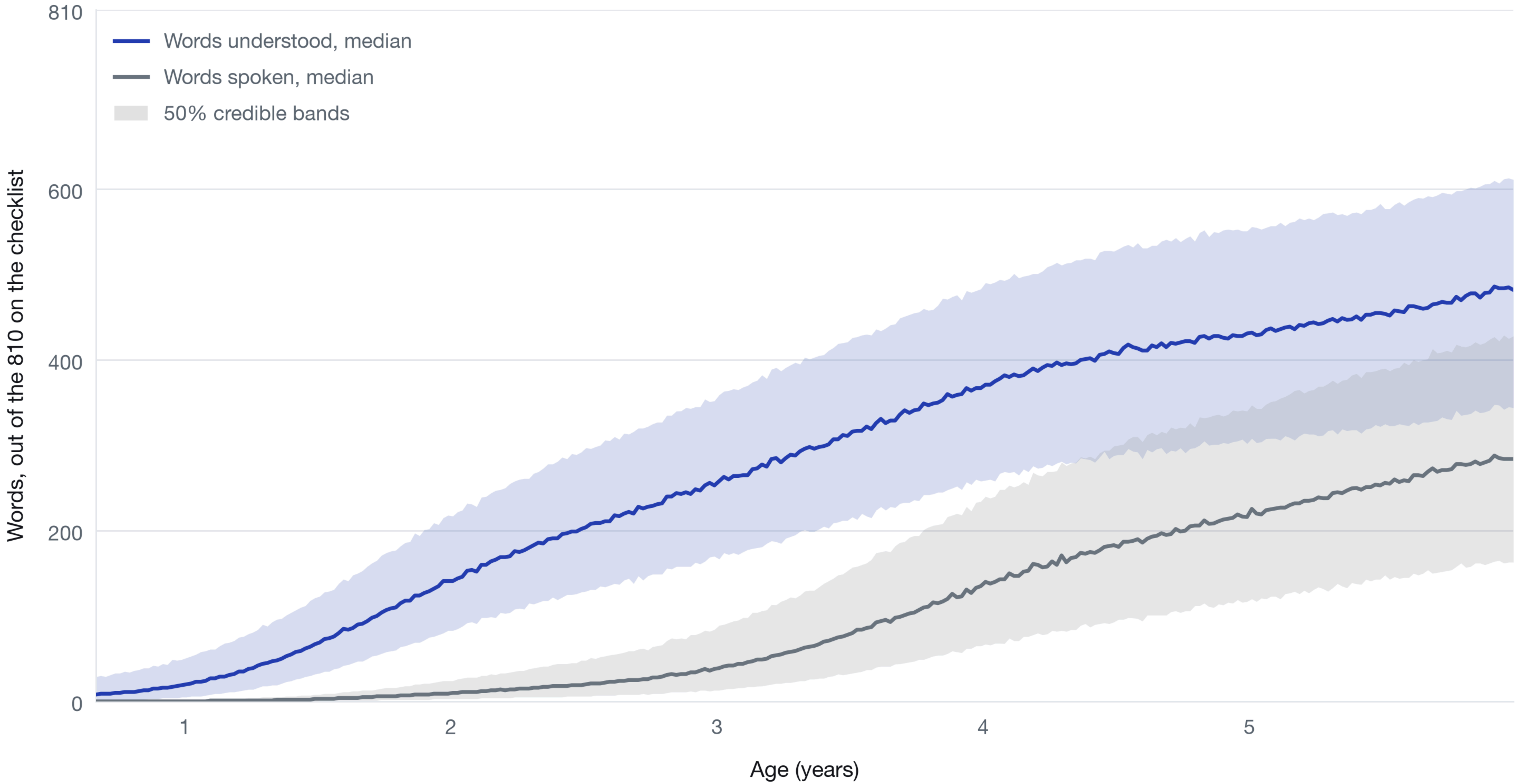
INSIDE: the model · the production ratio · findings

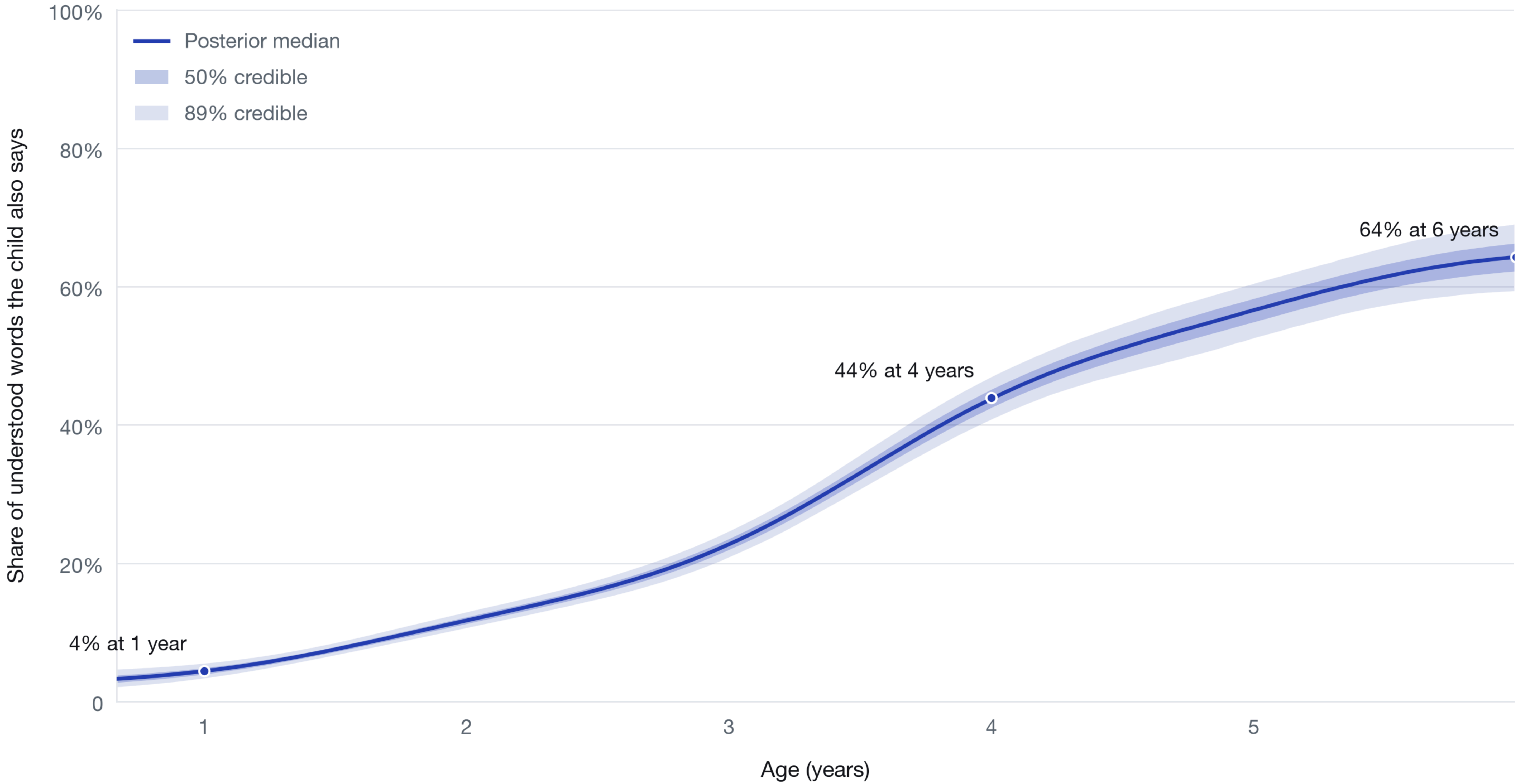
Words understood and words spoken by age, children with Down syndrome

Posterior-predictive medians with 50% credible bands. The model estimates spoken vocabulary as words understood multiplied by the share also said, so speaking can never exceed understanding, and the gap between the curves is the comprehension-production gap.



Share of understood words that children with Down syndrome also say, by age

Posterior median with 50% and 89% credible bands, from a hierarchical Bayesian model fitted to 1,424 parent-report vocabulary assessments pooled from 14 datasets in six countries.



What the study finds

The four headline findings from the pooled fit.

- **Understanding runs years ahead of speaking**
  The production ratio is the share of understood words a child also says. It climbs steeply with age, reaching about 44% by age four and 64% by six, the oldest age the model reports.

- **Spoken vocabulary lags furthest**
  By around age two, most children with Down syndrome say fewer words than the lowest-scoring tenth of typically-developing children. Understanding is delayed too, but by less.

- **The gap is wider than in typical development, then narrows**
  At the same level of comprehension, children with Down syndrome say a smaller share of the words they know. The difference peaks at around 13 percentage points at 50–150 understood words, then fades as vocabulary grows.

- **Signing offers a modest early bridge**
  Counting signed words credits children with expressive vocabulary they already have, and narrows the early gap somewhat. The narrowing is concentrated around ages two to three; the spoken-word gap stays large.

> The findings are preliminary and observational; estimates come from development-tier model fits and will sharpen as more data arrive.